

Amirhossein Masoudnezhad

Software Developer

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Profile

Software developer with a background in Physics and Electrical Engineering and experience building software for technical and scientific environments. Developed ClusterPilot, an iOS application for managing HPC clusters, with functionality for job monitoring, cluster file management, and batch script configuration. Comfortable working across mobile development, APIs, backend services, and technical computing environments.

Technical Skills

Mobile	Swift, SwiftUI, application architecture, UI development
Web & Backend	PHP, HTML/CSS, JavaScript, SQL, MongoDB, REST APIs
Programming	C/C++, Python, Dart/Flutter, Fortran, Bash
Systems & HPC	Git, Linux, HPC environments, MPI

Development Projects

ClusterPilot — Mobile HPC Management Platform —

- Designed and developed an iOS application for managing university and research HPC clusters.
- Implemented mobile access to SLURM job monitoring and submission, cluster file management, and SBATCH script configuration.
- Developed a centralized API layer connecting the mobile application with remote cluster services.

Scientific Software Development —

- Developed software and automated data-analysis pipelines for scientific applications.
- Worked with Python, C/C++, numerical methods, and data-processing workflows.

Education

MSc Astrophysics & Cosmology University of Bologna, Italy	2024–Present
BSc Physics Sharif University of Technology, Iran	2019–2022
BSc Electrical Engineering Sharif University of Technology, Iran	2016–2019

Selected Technical Experience

Zoomed Initial Conditions (ZInCo) — University of Bologna

- Extended and tested scientific software for generating multi-resolution cosmological initial conditions.
- Worked on particle management, numerical algorithms, resolution structures, and computational validation.

Machine-Learning Classification of Dusty Stellar Sources — IPM

- Developed automated data-analysis and machine-learning pipelines for astronomical observations.
- Worked with scientific datasets and computational workflows.

Additional Information

Languages: English (Fluent), Italian (Basic)